
International Invitational Math Optimization Challenge 2026: Solutions

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<https://github.com/welu2027/IIMOC-2026>

Abstract

This report presents my solutions to the two scored optimization problems of IIMOC 2026 (the International Invitational Math Optimization Challenge), *Rectangle Approximation* (99.19%) and *Chromatic Block Harmony* (99.92%), which together placed 8th of 307 overall. For Rectangle Approximation, I decompose the instance into connected components, place rectangles greedily with a zero-excess primitive derived from the largest-rectangle-in-a-histogram algorithm, allocate the output budget across components by marginal gain in the spirit of greedy submodular maximization, and refine with a coordinate-descent local search over rectangle edges. The local search alone recovers most of the gap to the top of the leaderboard. For Chromatic Block Harmony, I show that colors can be identified with block sums, reducing the problem to selecting at most B sums followed by an interval scheduling step that is solved exactly by the classical earliest-finish-time greedy. Sum selection then proceeds in three phases: an exact single-element baseline, incremental greedy addition, and budgeted swap refinement. I report ablations quantifying each component's contribution on the official judge distribution.

1 Introduction

Heuristic programming contests pose optimization problems with no efficiently computable optimum: any feasible output is accepted, and submissions are ranked by an objective evaluated on hidden test cases. Success depends less on a single clever idea than on a pipeline of mutually reinforcing components, each validated against the judge distribution. This report documents such pipelines for the two problems of the 2026 International Invitational Math Optimization Challenge (IIMOC).

The first problem, *Rectangle Approximation*, asks for K axis-aligned rectangles whose union best approximates the union of N given rectangles under symmetric-difference area. It is a geometric compression task closely related to covering rectilinear polygons with rectangles, which is NP-hard in general [4]. The second, *Chromatic Block Harmony*, asks for a maximum-profit selection of disjoint subarray blocks colored under an equal-sum constraint, and turns out to reduce cleanly to interval scheduling [3].

The contributions of this report are:

- a decomposition-plus-budgeting architecture for Rectangle Approximation that treats connected components as independent subproblems and allocates the global budget K by marginal gain, following the greedy paradigm for submodular maximization [5];
- an empirical comparison of two greedy placement primitives, showing that a restricted zero-excess rule based on the maximal-rectangle algorithm [7] dominates the more expressive max-gain rule based on the two-dimensional maximum-subarray problem [2] in both speed and final score;
- an edge-sliding local search, a coordinate descent [8] over the four edges of each output rectangle with exact $O(1)$ -per-cell move evaluation, worth +0.011 average score;

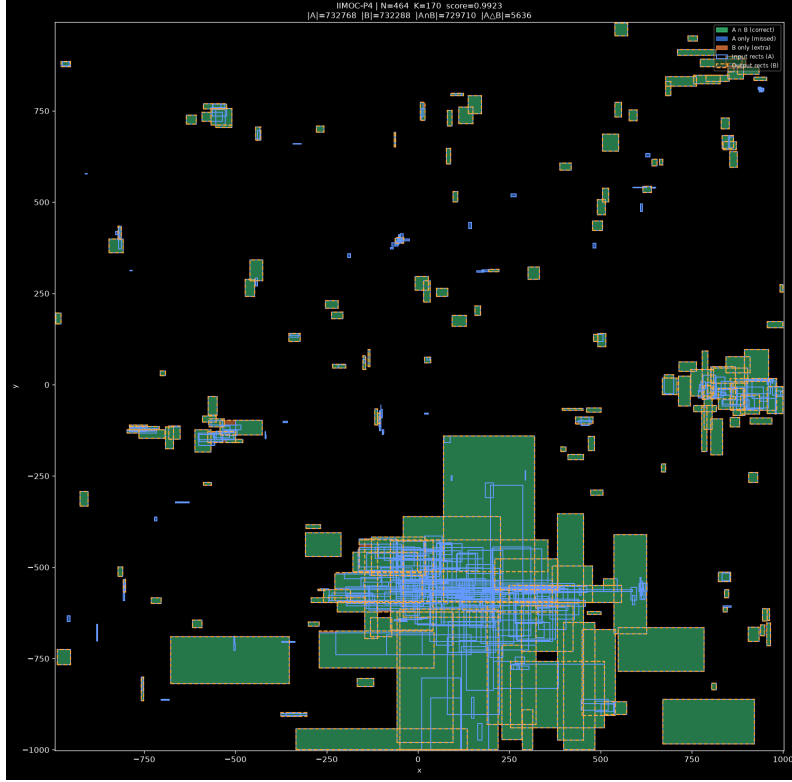


Figure 1: Judge case 7 ($N=464$, $K=170$, score 0.9923). Green: area covered by both A and B ; blue: area of A that was missed; orange: excess area outside A . The instance is typical of the generator: many isolated small rectangles, which are essentially free since each receives its own output rectangle, plus one dense clump where the error concentrates. The decomposition of Section 2.2 is what lets the budget allocator detect this structure and spend most of K on the clump.

- a reduction of Chromatic Block Harmony to candidate-sum selection, with an exactness argument for the packing step and an incremental-merge scheme that makes greedy selection affordable at $n = 50000$.

2 Rectangle Approximation

2.1 Problem statement

Given $N \leq 1000$ axis-aligned rectangles with integer coordinates in $[-1000, 1000]$, possibly overlapping, and a budget $K \leq N$, output exactly K axis-aligned integer rectangles. Let A be the union of the input and B the union of the output. The score is

$$\text{score} = \max\left(0, 1 - \frac{|A \Delta B|}{|A|}\right), \quad |A \Delta B| = |A| + |B| - 2|A \cap B|, \quad (1)$$

with areas measured in unit squares, under a 5 s time limit. The judge generator grows rectangles by iterated mutation of existing ones, yielding organic clumps, and draws K from $[C, \min(N, C + \max(1, \lfloor N/5 \rfloor))]$, where C is the number of connected clusters. The budget therefore always suffices to touch every cluster but is typically far below N , so overlapping rectangles must be merged into fewer, better ones.

2.2 Architecture

Decomposition. Rectangles whose interiors do not intersect, directly or transitively, cannot influence one another's contribution to the score. A union-find pass [6] therefore partitions the input into

connected components that are solved independently. Because the generator produces many small clumps rather than one large blob (Figure 1), this decomposition is also the main source of speed: each component’s grid is tiny compared to the full instance.

Coordinate compression. Only coordinates that appear in the input are relevant, so all grids are compressed: one cell represents a full axis-aligned box of real area, and cell widths and heights are carried explicitly throughout. A component with m rectangles yields a grid of at most $(2m - 1) \times (2m - 1)$ cells regardless of its geometric extent.

Greedy placement. Within a component, output rectangles are placed one at a time, each maximizing a value function. I implemented and compared two value functions:

1. **Max gain.** A candidate’s value is the new area of A it covers minus the area outside A it wastes. Assigning $+\text{area}$ to uncovered cells inside A and $-\text{area}$ to cells outside A makes this exactly the maximum-sum subrectangle problem, solvable in $O(n_x^2 n_y)$ by the standard reduction of the two-dimensional case to Kadane’s one-dimensional scan [2].
2. **Zero excess.** Candidates are restricted to rectangles lying entirely inside the uncovered part of A , so the value is plain area. The best such rectangle is found by the stack-based largest-rectangle-under-a-histogram sweep in $O(n_x n_y)$ [7].

The zero-excess rule dominated: roughly $25\times$ faster per call (8–10 ms versus 200–250 ms on typical components) and higher final score. The intuition is that a max-gain rectangle that swallows excess distorts the residual problem seen by every subsequent placement, whereas zero-excess placements leave a clean residual and defer boundary decisions to the local search of Section 2.3, which evaluates them exactly. The max-gain primitive is retained only for small instances solved globally without decomposition.

Budget allocation. Placements within a component come with a decreasing sequence of gains, so allocating K across components reduces to repeatedly granting the next rectangle to the component with the largest pending marginal gain, the same rule that underlies the classical greedy for monotone submodular maximization [5]. A max-heap of next gains, popped K times, implements this in $O((N + K) \log C)$.

One design choice matters disproportionately: every component starts at budget *zero*. Granting each component one rectangle by default appears safe, since an uncovered component contributes its entire area to the error, but when K is close to the cluster count C the floor starves the large components. Allowing a large component’s second and third rectangles to outbid a small component’s first was worth $+0.0245$ average score, the largest single improvement from any architectural decision in the solution.

Reverse direction. A complementary constructor, drop-down, starts from all m input rectangles of a component and repeatedly deletes the rectangle whose removal costs the least symmetric-difference area until k remain. It is preferable when k/m is high and the inputs already approximate themselves well. Both constructors run and the better-scoring result seeds the local search.

2.3 Local search

Constructive greedy leaves a characteristic artifact: seams of uncovered area between adjacent placements, visible as blue channels in Figure 2 (left), because zero-excess rectangles never overlap the boundary of what is already covered. The refinement stage removes these seams with two alternating moves, run as coordinate descent [8] until the component’s time slice expires. Since both moves apply only strictly improving changes, the objective is monotone and additional rounds are always safe.

Edge sliding. For each output rectangle, each of its four edges is optimized in turn while the other three stay fixed (Algorithm 1). Sliding an edge outward annexes a strip of cells; sliding it inward releases one. Move values are computed exactly from two per-cell arrays: $\text{signed}[c]$, equal to $+\text{area}$ if cell c lies inside A and $-\text{area}$ otherwise, and $\text{coverage}[c]$, the number of output rectangles containing c . Annexing a cell with coverage 0 changes the error by $-\text{signed}[c]$; releasing a cell

Table 1: Score progression on the judge distribution. The cumulative +3.94% over the baseline corresponds to most of the leaderboard interval between mid-pack and the top ten.

Configuration	Mean score
Baseline greedy	0.9326
Corrected placement gains	0.9267
Marginal-gain budget allocation from zero	0.9458
+ edge sliding and relocation	0.9693

with coverage 1 changes it by $+\text{signed}[c]$; cells shared with another rectangle contribute nothing in either direction. A full scan of one edge’s positions is therefore a single prefix sweep over strips. Edge sliding alone was worth about +0.011 average score, the largest algorithmic improvement in the solution.

Algorithm 1 Edge sliding for one output rectangle $R = [i_1, i_2) \times [j_1, j_2)$

```

1: for each edge  $e$  of  $R$  (left, right, bottom, top) do
2:    $\Delta^* \leftarrow 0$ ;  $p^* \leftarrow$  current position of  $e$ 
3:    $\Delta \leftarrow 0$ 
4:   for each position  $p$  moving  $e$  outward, then each  $p$  moving  $e$  inward do
5:      $s \leftarrow$  strip of cells swept between consecutive positions
6:     if outward then  $\Delta \leftarrow \Delta - \sum_{c \in s, \text{coverage}[c]=0} \text{signed}[c]$ 
7:     else  $\Delta \leftarrow \Delta + \sum_{c \in s, \text{coverage}[c]=1} \text{signed}[c]$ 
8:     end if
9:     if  $\Delta < \Delta^*$  then  $\Delta^* \leftarrow \Delta$ ;  $p^* \leftarrow p$ 
10:    end if
11:  end for
12:  if  $\Delta^* < 0$  then move  $e$  to  $p^*$  and update coverage
13:  end if
14: end for
```

Relocation. Edge sliding cannot escape a poor initial placement, so a second move deletes the output rectangle covering the least exclusive area, recomputes the best zero-excess rectangle in the freed space, and keeps the swap only if the new rectangle is strictly more valuable. This repairs components where the greedy constructor boxed itself in, and pairs naturally with edge sliding: relocation makes coarse jumps, sliding polishes.

2.4 Guards and special cases

If $K \geq N$, printing the input verbatim scores exactly 1, so this case is handled first. Sufficiently small instances skip decomposition and are solved globally with the max-gain primitive. Two work caps bound the size of any compressed grid the solver will rasterize; oversized components fall back to returning their largest input rectangles. The caps never trigger on the judge distribution, whose generator shrinks rectangles as N grows, but they bound the worst case against adversarial dense inputs. Per-component time slices are allotted in proportion to each component’s rectangle budget, which keeps one expensive component from consuming the refinement time of all others.

2.5 Evaluation

All measurements use the official generator (seed 42, cases 1–30), which is the judge distribution itself. Table 1 traces the score as components were added; Table 2 isolates the local search on four representative cases, and Figure 3 shows the full pipeline on two further cases. Averaged over all 30 cases the mean score is 0.9693 and the slowest case runs in 1.31 s against a 5 s limit. The residual low scorers are cases where K is small relative to the spatial spread of the input, so the budget itself is the binding constraint.

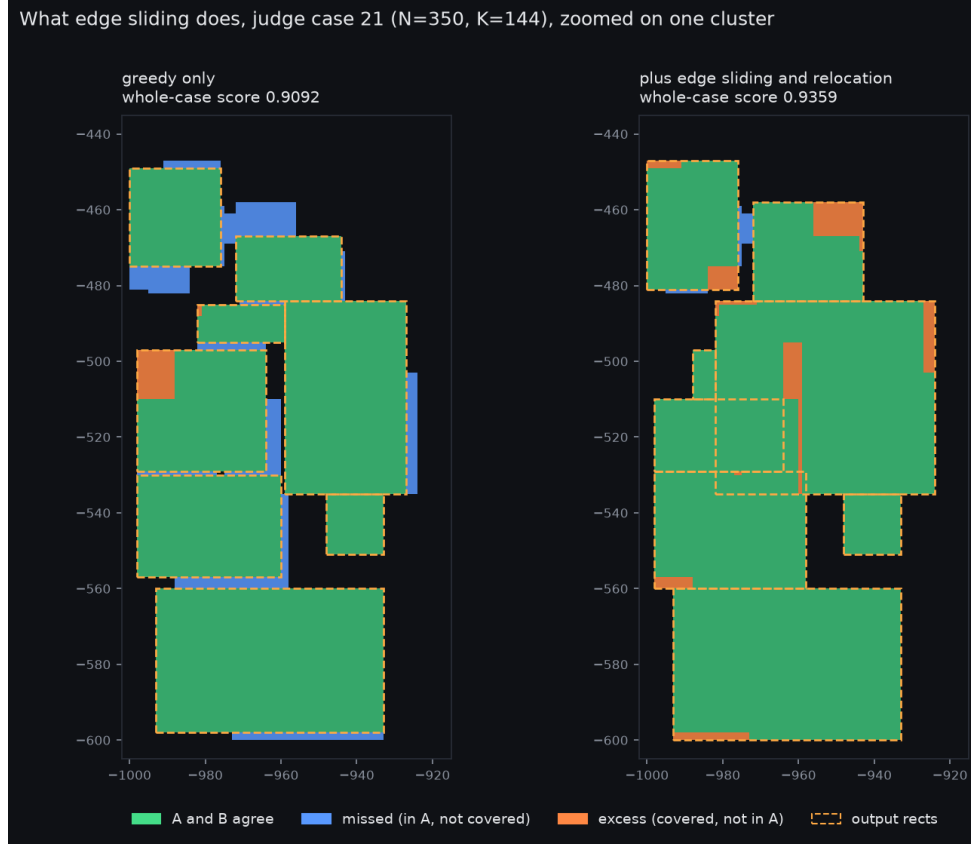
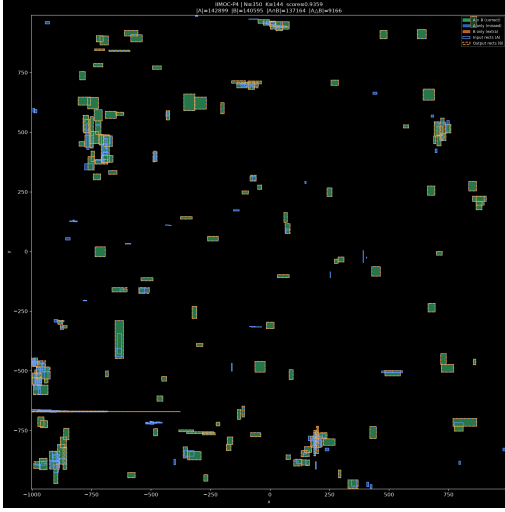


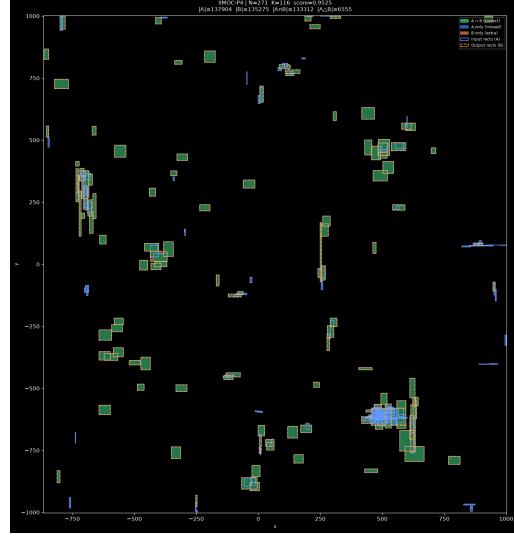
Figure 2: One component of judge case 21 without (left) and with (right) the local search of Section 2.3. Greedy zero-excess construction leaves blue seams between placements; edge sliding grows the rectangles across the seams, accepting a few slivers of orange excess in exchange for substantially more coverage. Only score-improving moves are kept.

Table 2: Local-search ablation on four judge cases.

Case	Constructive only	Full pipeline	Gain
5	0.9348	0.9525	+0.0177
21	0.9092	0.9359	+0.0267
24	0.9768	0.9841	+0.0073
27	0.9848	0.9901	+0.0053

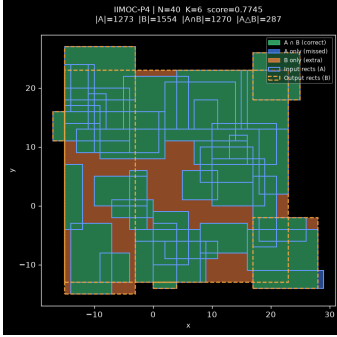


(a) Case 21 ($N=350$, $K=144$)

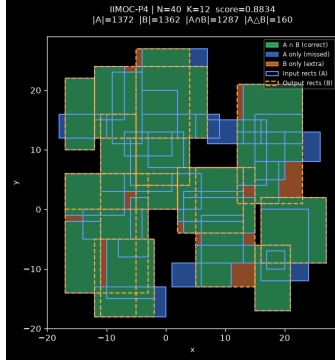


(b) Case 5

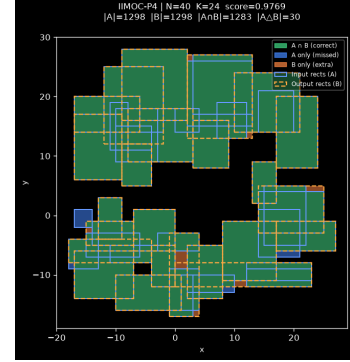
Figure 3: Two further judge cases after the full pipeline. Case 21 is one of the lower-scoring instances in Table 2: the input is spread across many small, far-apart clusters, so the connected-component decomposition of Section 2.2 produces many independent subproblems and K is spent thinly. Case 5 shows a denser single clump where local search fully closes the seams from greedy construction.



(a) $K = 6$



(b) $K = 12$



(c) $K = 24$

Figure 4: The same 40-rectangle input solved at three budgets.

Figure 4 illustrates how the budget shapes the solution on a hand-constructed 40-rectangle blob. At $K = 6$ the solver covers the blob with a few large rectangles and accepts heavy excess; at $K = 24$ it traces the boundary nearly exactly. Hand-made instances are used here because the generator’s small cases draw K close to N and are solved perfectly, which makes for an uninformative picture.

3 Chromatic Block Harmony

3.1 Problem statement

Given an array $a[1..n]$ of integers in $[0, 50]$ with $n \leq 50000$, and a color budget $B \leq 20$, select a set of pairwise disjoint blocks (contiguous subarrays) and assign each a color in $[1, B]$ such that all blocks of one color have equal sum. The objective is

$$\text{OBJ} = \alpha k - \beta C, \quad (2)$$

where k is the number of blocks, C the number of colors used, $\alpha \leq 10^9$ the per-block reward, and $\beta \leq 5 \times 10^8$ the per-color penalty. Selecting nothing scores 0, which is sometimes optimal. The time limit is 3 s.

3.2 Reduction to sum selection

The equal-sum constraint makes the color labels themselves immaterial: any feasible coloring can be relabeled so that each color corresponds to a distinct sum value, and conversely any assignment of blocks to at most B sums is feasible. The problem is therefore equivalent to:

choose a set S of at most B sums, then select a maximum number of pairwise disjoint blocks whose sums lie in S .

The inner packing problem is interval scheduling, for which the earliest-finish-time greedy is exactly optimal [3]: sort all candidate blocks by right endpoint and take each block that does not overlap the last one taken. All approximation in the solution thus lives in the choice of S ; given S , the packing is solved to optimality in linear time over the candidate blocks.

Blocks of a fixed sum s are enumerated with prefix sums: a block starting at L with sum s ends at the position r where $P[r] = P[L - 1] + s$, found by binary search. Since $a[i] \geq 0$, P is non-decreasing but not strictly increasing, so ties are resolved toward the shortest block ending at each r , which is the block most useful to the scheduler. Enumerations are cached per sum.

3.3 Selection in three phases

Candidate generation. Every value occurring in the array is a candidate, since single elements are valid blocks. Sums of windows of length 2 to 20 that recur sufficiently often are added. Candidates are ranked by a linear-time estimate of how many disjoint blocks each sum alone supports, and roughly the top 70 survive.

Phase 1: exact single-element baseline. Restricted to single-element blocks, disjointness is automatic and the objective decomposes as $\alpha \sum_{v \in S} \text{freq}(v) - \beta |S|$, which a frequency-sorted greedy optimizes exactly. This yields an instant lower bound that is frequently optimal outright when β is large relative to α .

Phase 2: incremental greedy addition. Sums are added to S one at a time, each addition chosen to maximize the resulting objective. The enabling data structure is an incrementally maintained merge of the chosen sums' block lists, kept sorted by right endpoint: evaluating a candidate is then one linear merge and one linear scheduling sweep rather than a rebuild, mirroring the lazy-evaluation economics that make greedy selection practical at scale.

Phase 3: budgeted swap refinement. Each chosen sum is tentatively exchanged for each unchosen candidate, a standard swap neighborhood from local search [1]. Prefix and suffix merges of the chosen lists are precomputed so that removing the i -th sum costs one merge rather than B . The phase runs under a 2.4 s budget against the 3 s limit and terminates as soon as a full pass finds no improving swap.

4 Discussion

Two findings generalize beyond these problems. First, the weaker placement primitive won. The max-gain rule is strictly more expressive than the zero-excess rule, yet produced worse pipelines, because a greedy step that optimizes its own local trade-off distorts the residual problem faced by every later step. Restricting construction to conservative moves and deferring all boundary trade-offs to a local search that evaluates them exactly proved both faster and more accurate. This mirrors the common pattern in combinatorial optimization of pairing a cheap constructive heuristic with a strong improvement phase [1].

Second, components must be measured in the pipeline, not in isolation. A merging pass that demonstrably improved individual components was ultimately removed because its runtime displaced

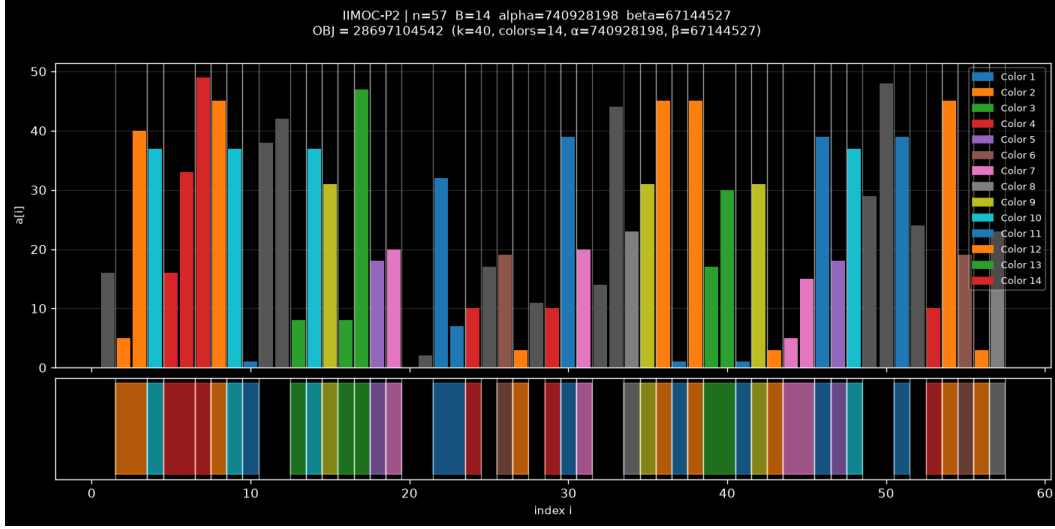


Figure 5: Judge case 28. Top: the array with selected blocks colored by assigned color; grey bars are unused elements. Bottom: the blocks as intervals. Blocks never overlap, and all blocks of one color share the same sum.

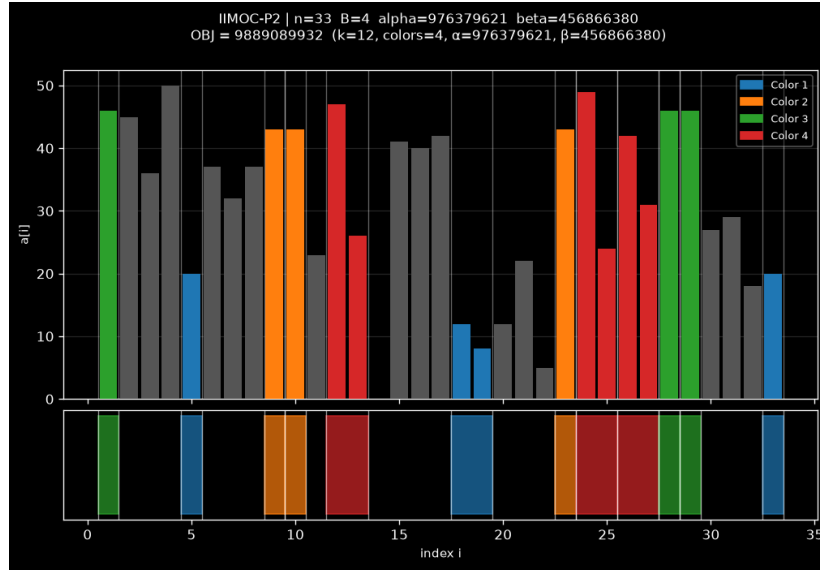


Figure 6: A small- n judge case in which multi-element blocks matter: several colors mix single elements with longer windows that reach the same sum.

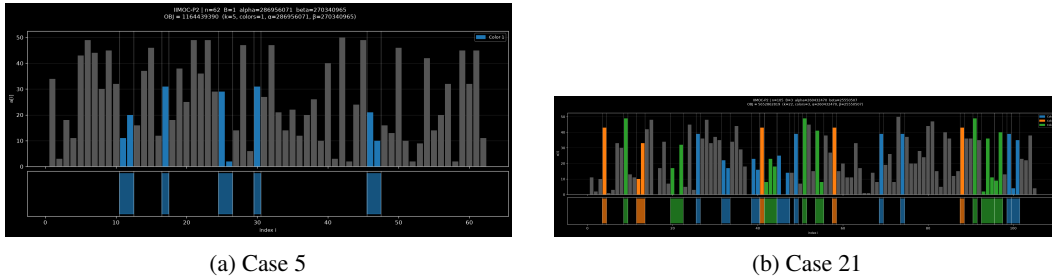


Figure 7: Two further small- n cases, chosen so the coloring stays legible. Large judge cases render as strips tens of thousands of pixels wide and are not reproduced here; only the small- n cases and the cases above are legible enough to inspect by eye.

refinement time elsewhere and lowered the end-to-end average. Under a shared time budget, the unit of evaluation is the full system on the judge distribution; per-component improvements are only proposals.

5 Reproducibility

All code, problem packages, and figures in this report are available at <https://github.com/welu2027/IIM0C-2026>. Both solutions are single Python files reading stdin and writing stdout:

```
python3 square_approximation.py < squareapproximation/smalltest/1.in
python3 block_harmony.py < blockharmony/smalltest/1.in
python3 gen.py 42 <case_id> # regenerate any judge case, ids 1..30
```

All figures were produced by feeding the solutions' outputs to the contest's official `vis.py`. In the rectangle plots, green is area covered by both A and B , blue is area of A that was missed, and orange is covered area outside A ; the score equals one minus (blue + orange) over $|A|$.

References

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